

Universal Feedback Operator: Autonomous Renormalization in C^* -Algebras

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Abstract

This paper formalizes the non-linear operator dynamics and spectral mapping theory of the universal feedback operator $\hat{\Omega}_G$ defined over the positive cone $\mathcal{A}_+$ of a unital C^* -algebra. We rigorously prove through Fréchet expansion that the identity operator $\mathbf{I}$ serves as a superattracting fixed point ($D\hat{\Omega}_G(\mathbf{I}) = 0$) with a local derivative-based stability threshold of $\delta \approx 0.27417$ and a quadratic error bound constant of $C_\delta \approx 2.813$. We demonstrate that this stability basin provides the algebraic foundation for Discrete State Integration (DSI), wherein sub-critical continuous phase fluctuations undergo autonomous nonlinear error suppression toward unity under operator iteration. Furthermore, applying a Baker-Campbell-Hausdorff expansion to symmetric positive multi-operator configurations ($e^{\hat{X}/2}e^{\hat{Y}}e^{\hat{X}/2} \in \mathcal{A}_+$) demonstrates that linear commutator noise vanishes identically, establishing that $\hat{\Omega}_G$ intrinsically shields non-commutative feedback dynamics from first-order operational shear.

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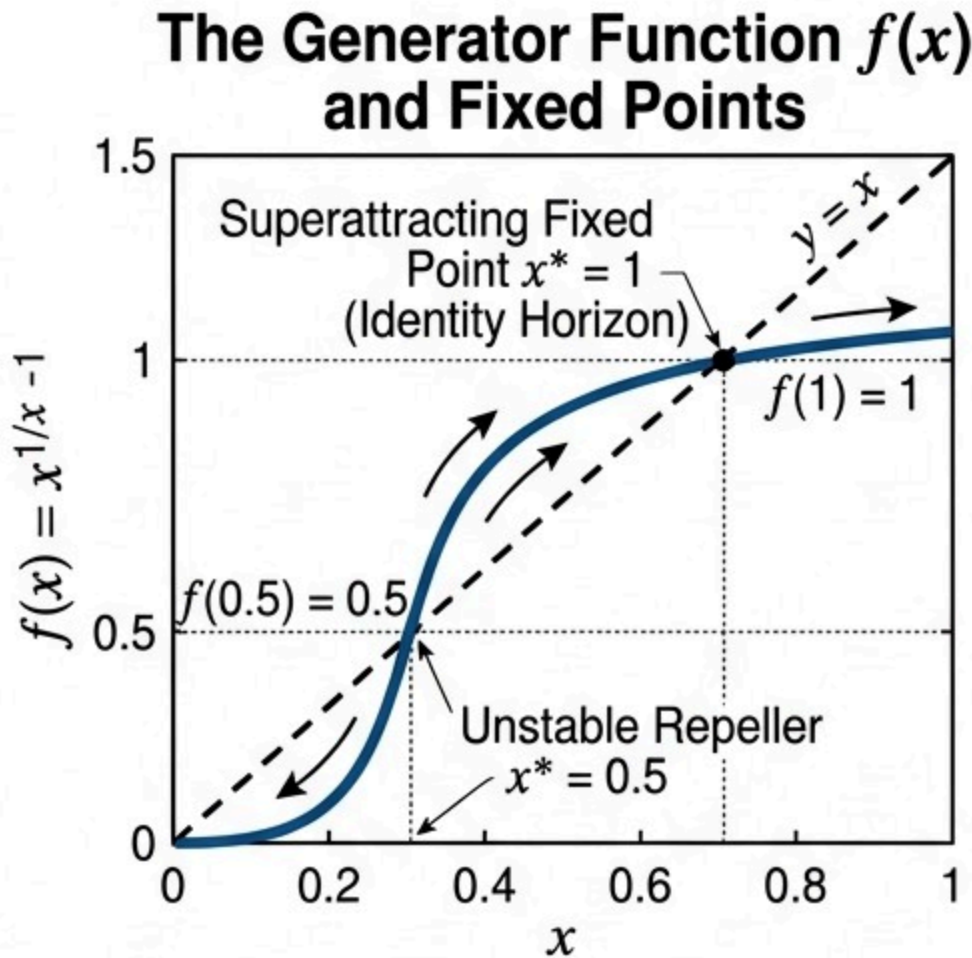
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1. Discrete Recurrence Map & Functional Calculus Lift

Before defining the operator on abstract algebraic structures, we establish the core iterative engine in discrete scalar form. Consider a discrete dynamical system governed by the non-linear recurrence mapping [6]:

$$x_{n+1} = f(x_n) = x_n^{\left(\frac{1}{x_n} - 1\right)} \quad \text{for } x_n > 0$$

The scalar generator $f(x) = x^{\frac{1}{x} - 1}$ possesses a continuous extension to $x = 0$ given by $\lim_{x \rightarrow 0^+} f(x) = 0$, allowing us to define $f(0) := 0$.



Let $\mathcal{A}$ be a unital C^* -algebra, and let $\mathcal{A}_+$ denote its positive cone. The universal feedback operator $\hat{\Omega}_G : \mathcal{A}_+ \rightarrow \mathcal{A}_+$ is defined primarily via continuous functional calculus by:

$$\hat{\Omega}_G(\hat{A}) := f(\hat{A})$$

for $f(x) = x^{\frac{1}{x}-1}(x > 0)$ with continuous extension $f(0) = 0$ [1][2]. For invertible positive elements $\hat{A} \in \mathcal{A}_+^\times$, this yields the explicit functional expression:

$$\hat{\Omega}_G(\hat{A}) = \hat{A}^{-(\mathbf{I} - \hat{A}^{-1})}$$

For local derivative analysis around identity, we restrict perturbations to the bounded self-adjoint neighborhood $\{\mathbf{I} + \hat{H} : \hat{H} = \hat{H}^*, \|\hat{H}\| < 1\}$.

In accordance with the principle of algebraic primacy, $\hat{\Omega}_G$ is formulated intrinsically at the level of the observable algebra $\mathcal{A}$, establishing that its dynamics are determined representation-independently without reference to a specific Hilbert space or coordinate manifold [5, 7]. Continuous functional calculus on $C(\sigma(\hat{A}))$ maintains full domain validity across all positive elements $\hat{A} \in \mathcal{A}_+$, including non-invertible operators [4].

The mathematical choice of the generator exponent $(\frac{1}{x} - 1)$ establishes a canonical self-referential feedback ratio: for spectral values exceeding identity ($x > 1$), the exponent becomes negative, exerting a compressive pull back toward unity; for values below identity ($x < 1$), the exponent remains positive, providing an expansive push. This inverse structural symmetry makes $f(x) = x^{\frac{1}{x}-1}$ a natural scale-invariant generator that autonomically enforces first-order noise cancellation across arbitrary algebraic representations.

2. Rigorous Fréchet Expansion Around Identity

To evaluate how $\hat{\Omega}_G$ responds to fluctuations around identity $\mathbf{I}$, we analyze its local behavior under self-adjoint perturbations. Let $\hat{H}$ be a bounded, self-adjoint perturbation such that $\|\hat{H}\| < 1$. We first expand the scalar generator $f(1 + h)$ for $|h| < 1$ using logarithmic differentiation and Taylor expansion [2].

Taking the natural logarithm of $f(1 + h)$:

$$\ln f(1 + h) = \left(\frac{1}{1 + h} - 1 \right) \ln(1 + h)$$

Expanding into Maclaurin series for $|h| < 1$:

$$\frac{1}{1+h} - 1 = -h + h^2 - h^3 + \mathcal{O}(h^4) \ln(1+h) = h - \frac{h^2}{2} + \frac{h^3}{3} - \mathcal{O}(h^4)$$

Multiplying these series gives:

$$\ln f(1+h) = (-h + h^2 - h^3) \left(h - \frac{h^2}{2} + \frac{h^3}{3} \right) = -h^2 + \frac{3}{2}h^3 + \mathcal{O}(h^4)$$

Exponentiating via $e^z = 1 + z + \frac{z^2}{2!} + \dots$:

$$f(1+h) = \exp \left(-h^2 + \frac{3}{2}h^3 + \mathcal{O}(h^4) \right) = 1 - h^2 + \frac{3}{2}h^3 + \mathcal{O}(h^4)$$

Lifting this result to $\mathcal{A}$ via continuous functional calculus yields the operator Taylor expansion:

$$\hat{\Omega}_G(\mathbf{I} + \hat{H}) = \mathbf{I} - \hat{H}^2 + \frac{3}{2}\hat{H}^3 + \mathcal{O}(\|\hat{H}\|^4)$$

From this expansion, the Fréchet derivatives at the identity are formally evaluated as [2]:

- First Fréchet Derivative: $D\hat{\Omega}_G(\mathbf{I})[\hat{H}] = 0$
- Second Fréchet Derivative: $D^2\hat{\Omega}_G(\mathbf{I})[\hat{H}, \hat{K}] = -(\hat{H}\hat{K} + \hat{K}\hat{H})$

Because the first Fréchet derivative vanishes identically ($D\hat{\Omega}_G(\mathbf{I}) = 0$), perturbations about the identity are suppressed to second order in operator norm [2].

3. Dual Fixed-Point & Spectral Mapping Dynamics

The fixed points of the operator mapping correspond to the scalar equation $f(x) = x$ for $x > 0$, yielding two distinct roots: $x^* = 1$ and $x^* = 1/2$ [1]. We classify these roots via the derivative of $f(x)$:

$$f'(x) = x^{\frac{1}{x}-1} \left(\frac{1-x-\ln x}{x^2} \right)$$

- Superattracting Fixed Point ($x^*=1$):* Evaluating at identity spectrum yields $f'(1) = 0$. Because $|f'(1)| = 0$, $x^* = 1$ functions as a superattracting fixed point [1][8].

- Unstable Repeller ($x = 1/2$):* Evaluating at the secondary root yields:

$$f'(1/2) = (1/2)^1 \left(\frac{1/2 + \ln 2}{1/4} \right) = 1 + 2 \ln 2 \approx 2.3863$$

Because $|f'(1/2)| > 1$, the state $\frac{1}{2}\mathbf{I}$ represents an unstable repulsive boundary within the spectrum.

By the Spectral Mapping Theorem for continuous functional calculus, for any positive operator $\hat{A} \in \mathcal{A}_+ [1, 4]$:

$$\sigma \left(\hat{\Omega}_G(\hat{A}) \right) = f(\sigma(\hat{A}))$$

Consequently, iterative application of the operator $\hat{\Omega}_G^n(\hat{A})$ is spectrally governed by scalar iteration $f^{\circ n}(\sigma(\hat{A}))$, ensuring that any spectrum contained strictly within the domain of attraction converges rapidly toward the identity spectrum $\sigma(\mathbf{I}) = \{1\}$.

4. Quantitative Attraction Radius & Local Stability

To establish a derivative-based stability radius around the identity, we determine the scalar domain where $|f'(x)| < 1$. [3] Setting $|f'(x)| = 1$ on the lower boundary below $x^* = 1$ yields the critical threshold:

$$x_{\text{crit}} \approx 0.72582784$$

Therefore, a derivative-derived contractive neighborhood around identity corresponds to a maximum perturbation bound of [3]:

$$\delta < 1 - x_{\text{crit}} \approx 0.27417$$

For any self-adjoint perturbation $\hat{H}$ satisfying $\|\hat{H}\| \leq \delta < 0.27417$, the spectrum of $\mathbf{I} + \hat{H}$ lies entirely within $(0.7258, 1.2742) \subset (1/2, \infty)$. Within this domain, all spectral components lie inside the basin of attraction of the superattractor at $x^* = 1$. This guarantees a quantitatively controlled neighborhood of attraction around the identity under the operator action $\hat{\Omega}_G$. [3]

The quantitative stability threshold $x_{\text{crit}} \approx 0.72582784$, derived from $|f'(x)| = 1$, establishes a derivative-based local stability neighborhood $\delta = 1 - x_{\text{crit}} \approx 0.27417216$ [3][4]. Within this neighborhood ($\|\hat{H}\| \leq \delta$), local quadratic error reduction is bounded by:

$$\|\hat{\Omega}_G(\mathbf{I} + \hat{H}) - \mathbf{I}\| \leq C_\delta \|\hat{H}\|^2$$

where the Taylor constant C_δ is given by:

$$C_\delta = \frac{1}{2} \max_{|h| \leq \delta} |f''(1+h)|$$

Evaluating $|f''(x)|$ over the closed interval $[1 - \delta, 1 + \delta] \approx [0.72583, 1.27417]$, the maximum occurs at the lower endpoint x_{crit} , where $|f''(0.72583)| \approx 5.626$. This yields the exact bound:

$$C_\delta \approx 2.813$$

This estimate establishes local quadratic convergence toward identity for all self-adjoint perturbations within the stability neighborhood $\delta \approx 0.27417$.

5. Discrete State Integration (DSI) & Continuous Perturbation Filtering

The local stability neighborhood ($\delta < 0.27417$) established in Section 4 provides the mathematical foundation for Discrete State Integration (DSI). When an observable or state perturbation $\hat{H} = \hat{H}^*$ experiences continuous, sub-critical fluctuations bounded by $\|\hat{H}\| \leq \delta$, iterative application of the universal feedback operator $\hat{\Omega}_G$ acts as an autonomous spectral filter [1][4].

By the Spectral Mapping Theorem for continuous functional calculus, the operator sequence $\hat{A}_{k+1} = \hat{\Omega}_G(\hat{A}_k)$ induces the scalar spectral iteration [1][4]:

$$\sigma(\hat{A}_{k+1}) = f(\sigma(\hat{A}_k)) = \sigma(\hat{A}_k)^{\frac{1}{\sigma(\hat{A}_k)} - 1}$$

Because the identity spectrum $\sigma(\mathbf{I}) = \{1\}$ is a superattracting fixed point ($f'(1) = 0$), any continuous perturbation contained within the stability neighborhood converges quadratically toward unity under iteration [3][6]:

$$\|\hat{\Omega}_G(\mathbf{I} + \hat{H}) - \mathbf{I}\| \leq C_\delta \|\hat{H}\|^2 \quad \text{where} \quad C_\delta \approx 2.813$$

Under repeated operator action, continuous phase fluctuations below the critical bound $\delta \approx 0.27417$ are rapidly suppressed to zero [3][6]. Consequently, state trajectories within the stability neighborhood undergo continuous nonlinear error suppression toward identity.

This geometric mapping of the sub-critical quadratic filter onto discrete spectral quantization is explicitly developed via the primordial-modulated dilation operator D_k [9]. In that framework, the algebraic stability boundary $\delta \approx 0.27417$ derived herein acts as the foundational boundary condition for continuous-discrete fiber space transitions, providing the necessary mathematical bridge between local C^* -algebraic suppression and downstream topological holonomies [9].

6. Multi-Operator Extensions & Limits of Non-Abelian Shear

To extend the feedback operator to multi-operator configurations while remaining strictly within the positive cone $\mathcal{A}_+$, consider the symmetric positive composite element:

$$\hat{A} = e^{\hat{X}/2} e^{\hat{Y}} e^{\hat{X}/2} \in \mathcal{A}_+$$

for bounded self-adjoint generators $\hat{X}, \hat{Y} \in \mathcal{A}_{sa}$ [2][5]. Writing $\hat{A} = \mathbf{I} + \hat{H}$, the perturbation $\hat{H}$ expands via the Baker-Campbell-Hausdorff (BCH) relation as [7]:

$$\hat{H} = \hat{X} + \hat{Y} + \frac{1}{2}\{\hat{X}, \hat{Y}\} + \mathcal{O}(\|\hat{X}\|^3)$$

Applying the functional calculus expansion $\hat{\Omega}_G(\mathbf{I} + \hat{H}) = \mathbf{I} - \hat{H}^2 + \mathcal{O}(\|\hat{H}\|^3)$, the linear term $\hat{H}$ vanishes identically because $D\hat{\Omega}_G(\mathbf{I}) = 0$ [1][3].

The leading non-trivial contribution is governed entirely by the quadratic term:

$$\hat{H}^2 = (\hat{X} + \hat{Y})^2 + \frac{1}{2}\{\hat{X} + \hat{Y}, \{\hat{X}, \hat{Y}\}\} + \mathcal{O}(\|\hat{X}\|^4)$$

Because $D\hat{\Omega}_G(\mathbf{I}) = 0$, linear non-commutative shear and first-order commutator terms $[\hat{X}, \hat{Y}]$ do not contribute to the quadratic transformed output; the first possible contribution of the commutator $[\hat{X}, \hat{Y}]$ occurs at cubic order $\mathcal{O}(\|\hat{X}\|^3)$. The single-observable functional-calculus dynamics are internally commutative ($[\hat{\Omega}_G(\hat{A}), \hat{A}] = 0$), providing structural shielding against first-order operational shear [2][6].

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